DO NOW — RECALL

Lesson 139 · SQL INSERT, UPDATE and DELETE Safely

Name: _____ Class: _____ Date: _____

1. What is the difference between WHERE and ORDER BY in a query?

2. What are the outputs of a half adder when A=1 and B=1?

3. In Big O notation, what is binary search worst-case complexity?

4. When does a linear search stop?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

WHERE filters rows; ORDER BY arranges returned rows.

2. Model answer

Sum 0, carry 1.

3. Model answer

O(log n) for sorted random-access data.

4. Model answer

On a match or after examining all items.

By the end, you can…

• INSERT INTO Table(columns) VALUES(...).

• UPDATE Table SET column=value WHERE key=...

• DELETE FROM Table WHERE key=...; verify matches before writing.

Vocabulary — know these terms

INSERT — Adds new rows.

UPDATE — Changes values in existing rows.

DELETE — Removes matching rows.

WHERE — Restricts affected rows; omission can affect every row.

NEW LEARNING

Read, annotate and explain

Create rows

INSERT INTO Table(columns) VALUES(...).

Change rows safely

UPDATE Table SET column=value WHERE key=...

Delete deliberately

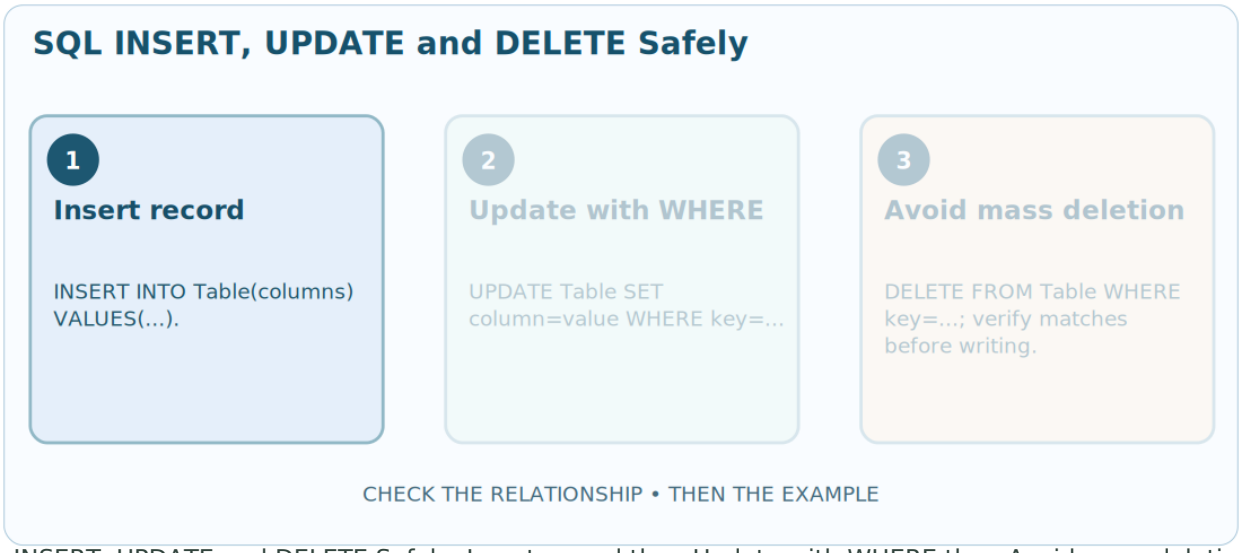
DELETE FROM Table WHERE key=...; verify matches before writing.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

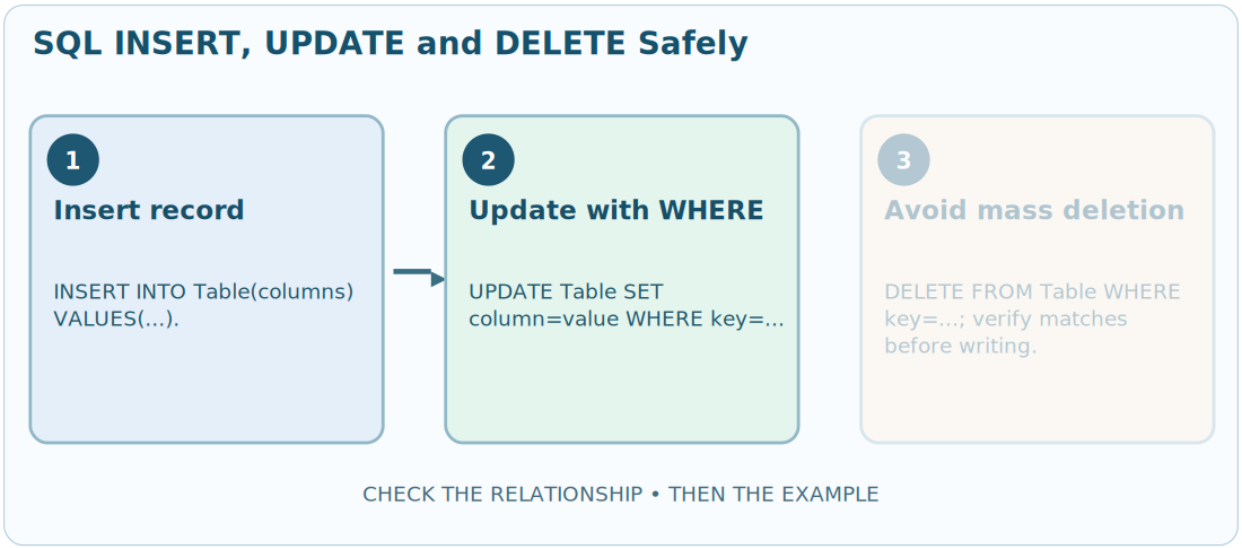
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



SQL INSERT, UPDATE and DELETE Safely; Insert record then Update with WHERE then Avoid mass deletion

New Learning

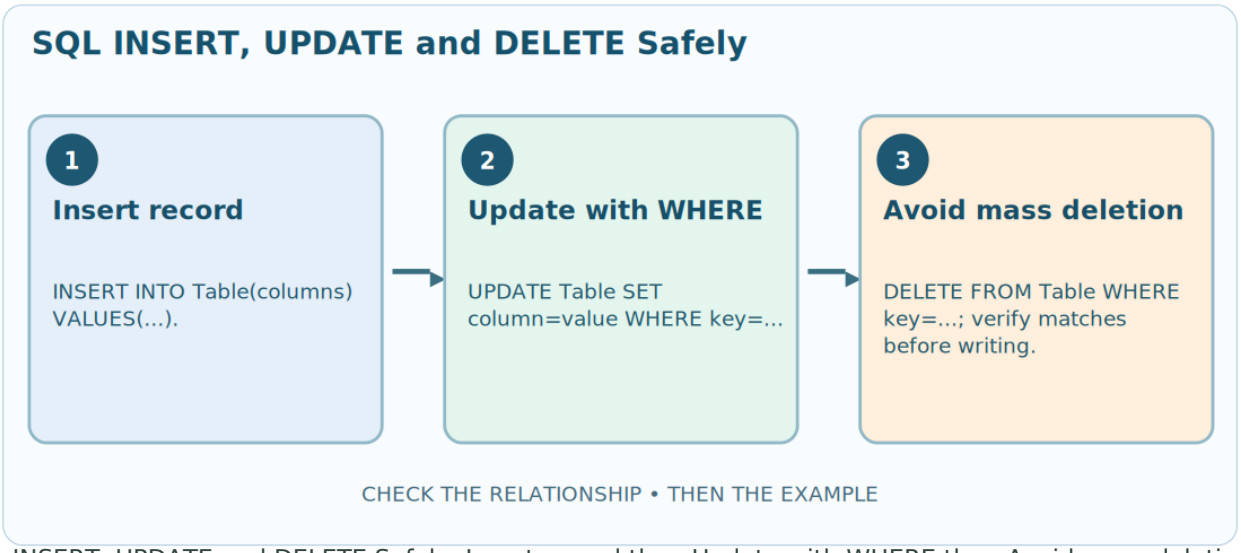


SQL INSERT, UPDATE and DELETE Safely; Insert record then Update with WHERE then Avoid mass deletion

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



SQL INSERT, UPDATE and DELETE Safely; Insert record then Update with WHERE then Avoid mass deletion

I DO — WORKED EXAMPLE

Follow the worked method

A Book table has BookID=8,Stock=4. Write an UPDATE to set Stock=3 for that row.

Identify BookID 8

Find the given data and name the required result.

UPDATE Stock = 3

Explain why this intermediate stage is necessary.

WHERE BookID = 8

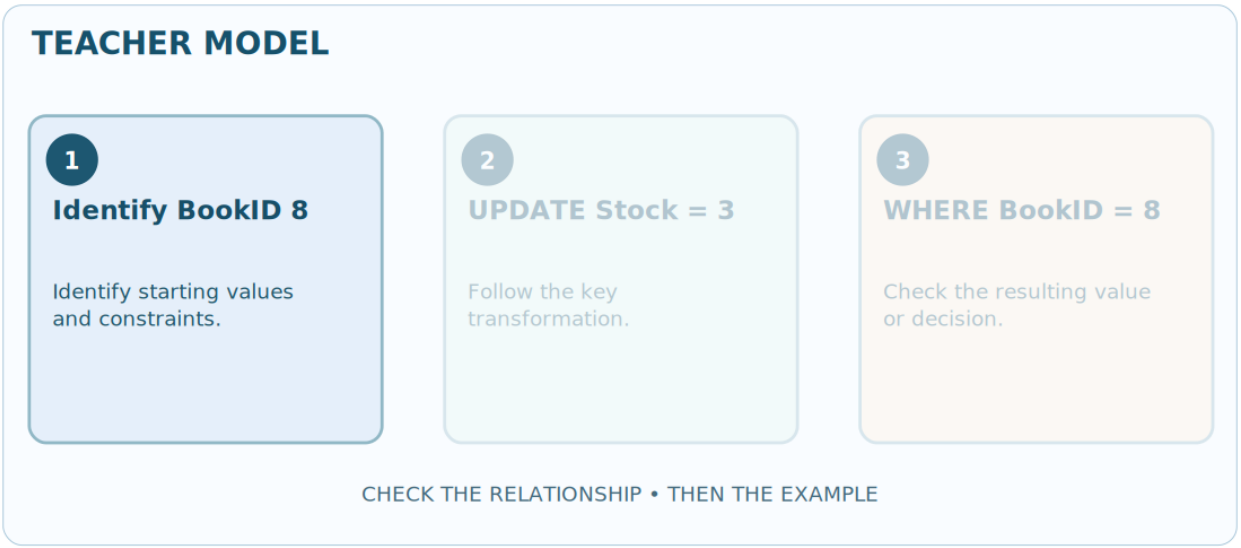
UPDATE Book SET Stock = 3 WHERE BookID = 8; Before running, SELECT the matching row. Without WHERE, the stock column of all books would be changed.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A Book table has BookID=8,Stock=4. Write an UPDATE to set Stock=3 for that row.

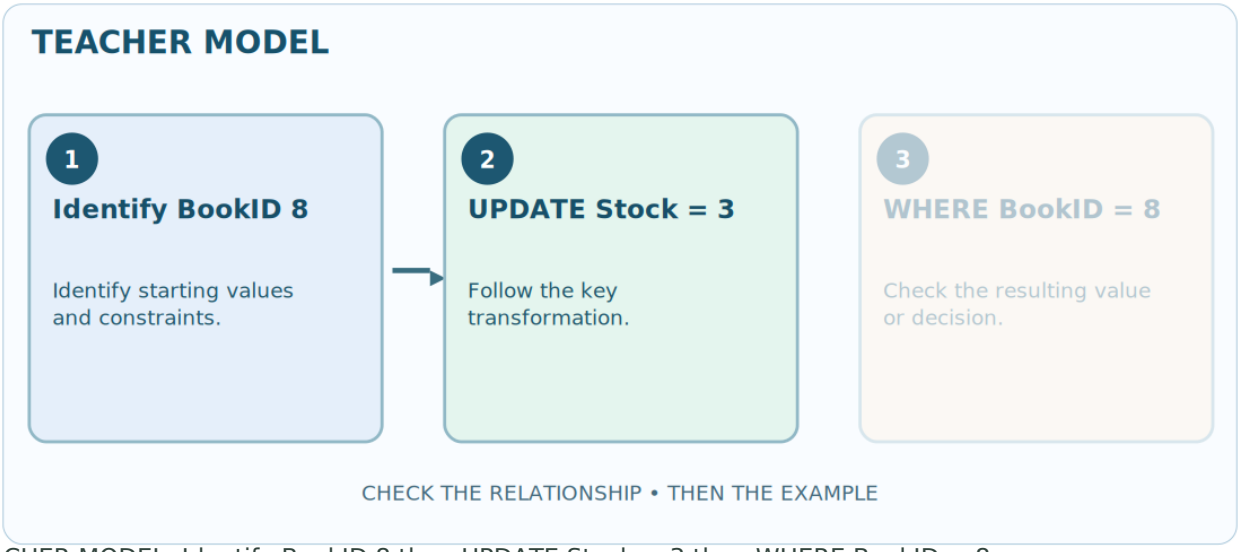
Find the given data and name the required result.



TEACHER MODEL; Identify BookID 8 then UPDATE Stock = 3 then WHERE BookID = 8

A Book table has BookID=8,Stock=4. Write an UPDATE to set Stock=3 for that row.

Find the given data and name the required result.



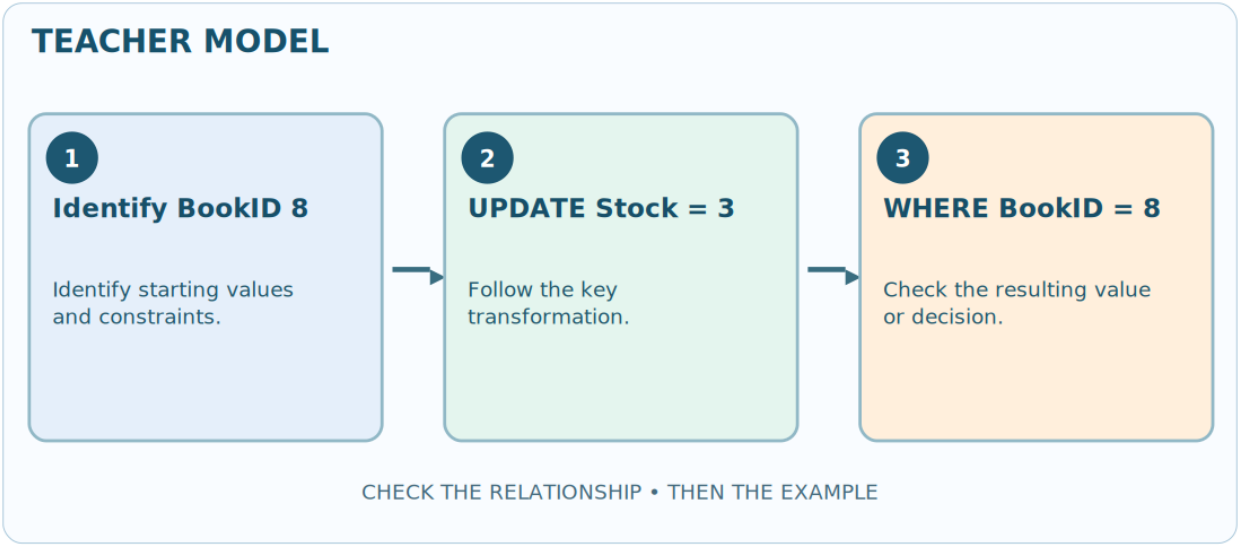
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I Do · original lesson diagrams

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TEACHER MODEL; Identify BookID 8 then UPDATE Stock = 3 then WHERE BookID = 8

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Add a new Club record and delete only the club with ClubID=12.

Find the given data and name the required result.



GUIDED PRACTICE; Insert club 24 then Confirm club 12 exists then DELETE WHERE ClubID=12

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Insert club 24 Find the given data and name the required result.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Write SQL inserting StudentID 22 and Name 'Kai' into Student. [4]

Question 2

A teacher must change a single pupil's YearGroup to 13. Explain a safe SQL statement. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain risks in DELETE FROM Booking; and suggest protections in a relational production database. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

UPDATE Book SET Stock=0; changes exactly one Book record.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

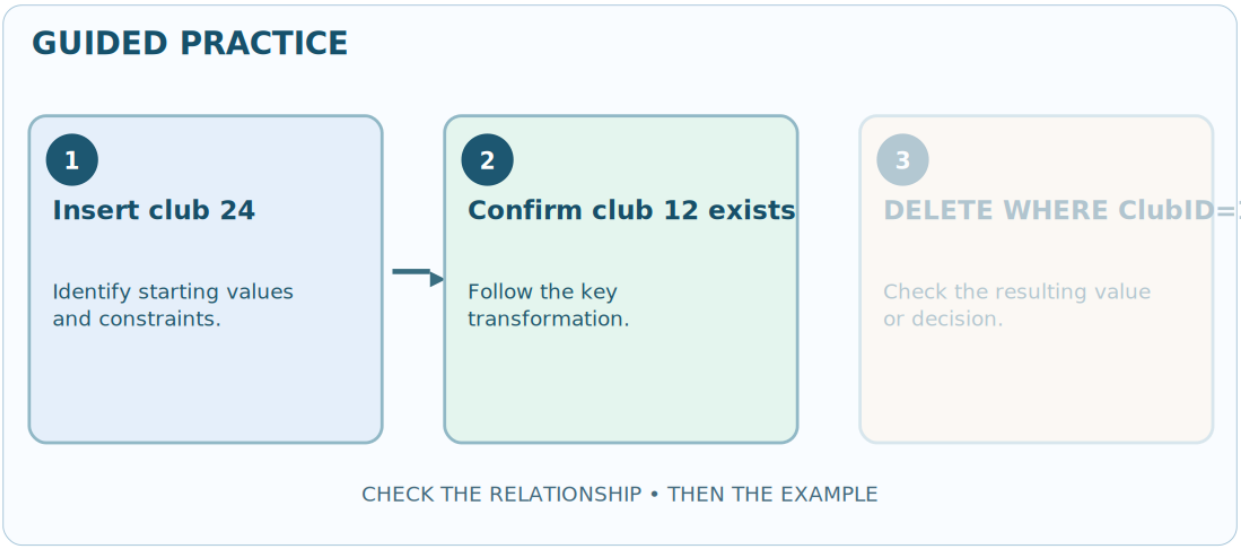
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Add a new Club record and delete only the club with ClubID=12.

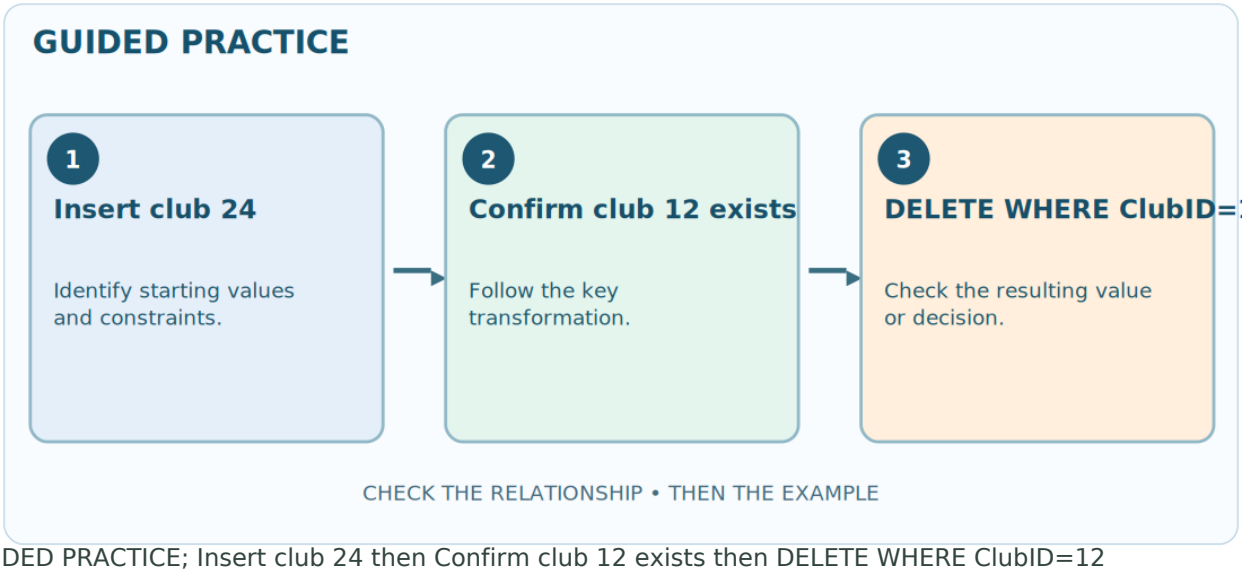
Find the given data and name the required result.



GUIDED PRACTICE; Insert club 24 then Confirm club 12 exists then DELETE WHERE ClubID=12

Add a new Club record and delete only the club with ClubID=12.

Find the given data and name the required result.



GUIDED PRACTICE; Insert club 24 then Confirm club 12 exists then DELETE WHERE ClubID=12

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Confirm club 12 exists

Explain why this intermediate stage is necessary.

DELETE WHERE ClubID=12

INSERT INTO Club (ClubID, Name) VALUES (24, 'Robotics'); DELETE FROM Club WHERE ClubID = 12; Foreign-key dependencies may prevent delete until related records are handled by defined policy.

IMPROVE: Explain one correction to your own reasoning.